DSANet: A Grid-Search Optimized Heterogeneous Ensemble for Ischemic Stroke Lesion Segmentation in Low-Resource Settings

Abstract. Accurate segmentation of ischemic stroke lesions from multi-modal MRI is a critical prerequisite for timely clinical decision-making, yet most state-of-the-art methods rely on heavy architectures and high-end GPUs that are impractical in low-resource clinical settings. This paper presents DSANet, a heterogeneous ensemble framework that couples three complementary 3D architectures, a densely connected network, a residual segmentation network, and an attention-gated U-Net—through a principled grid-search optimization pipeline. Rather than fusing model outputs with uniform or heuristic weights, we formulate ensemble construction as a constrained search problem over both fusion weights and the binarization threshold, with optimal values selected on the validation set via a Dice-driven objective. The pipeline further incorporates test-time augmentation on axis-flipped variants and morphological post-processing to suppress small disconnected false positives. Multi-modal information from FLAIR, DWI, and ADC sequences is integrated into a unified 3D input representation, and training is kept resource-efficient through automatic mixed precision, gradient accumulation, and cosine-annealed AdamW. Experiments on the ISLES 2022 dataset under a fixed 70/15/15 split show that DSANet attains a macro Dice Similarity Coefficient of 0.8780, a micro F1-score of 0.8780, and a recall of 0.9279, outperforming each constituent network and a DynUNet baseline by a clear margin. A fine-grained subgroup analysis confirms stable performance across lesion sizes (0.7306 on small infarcts and 0.8860 on large lesions) and anatomical regions (0.9541 in deep brain structures). These results demonstrate that carefully optimized heterogeneous ensembles offer a practical path to accurate stroke segmentation under constrained computational budgets. To support reproducibility and transparency, we provide a dedicated website hosting the full pipeline, implementation details, and experimental results available at Github .

Keywords: DSANet · Grid Search · Ischemic Stroke · Segmentation · ISLES

1 Introduction

Ischemic stroke [1] is one of the leading causes of disability and mortality worldwide. Rapid and accurate identification of the affected brain region is essential for effective treatment and prognosis. Automated stroke segmentation can assist clinicians by providing consistent and objective lesion maps, reducing diagnostic time and improving treatment planning. Multi-modal MRI sequences, such as FLAIR, diffusion-weighted imaging (DWI), and apparent diffusion coefficient (ADC), provide complementary information about tissue changes [1,2].

Existing methods have explored both traditional image processing techniques and deep learning approaches. Classical methods often rely on intensity thresholding or region-growing algorithms, which may fail in the presence of noise or subtle lesions. Kuang et al. [3] proposed LBMNet, a hybrid CNN–Mamba model, which received an 82% Dice score at ISLES 2022. However, these complex structures represent a large computational cost, and performance reliability may suffer across a wide range of clinical samples. Multi-model ensemble methods provide an essential role in this generalization. Garcia-Salgado et al. [4] present a deep learning-based MRI stroke lesion segmentation method. However, the proposed method has problems of generalization and adaptability. DeepISLES [5] is a powerful multi-modal ensemble-based stroke lesion segmentation method achieving high Dice scores on ISLES 2022. However, its high computational demands and GPU requirements make it challenging for practical use in low-resource systems where computing resources like high-end GPUs and huge memory systems are not easily available. Most existing models for stroke segmentation depend on complex architectures and high computational overhead, making them impractical for low-resource settings and limiting stroke diagnosis. As a result, it is critical to provide segmentation frameworks that are not only accurate but also computationally efficient, allowing deployment on restricted hardware without compromising speed.

Addressing these problems is critical for enhancing the reliability and accessibility of automated stroke segmentation. To address those challenges, this paper introduces DSANet-ISLES, a resource-efficient ensemble framework for accurate stroke lesion segmentation. The proposed DSANet-ISLES integrates multi-modal MRI inputs into a unified 3D segmentation framework to enhance diagnostic precision in resource-constrained environments. Instead of relying on a single complex model, the framework employs a set of complementary lightweight models to capture diverse spatial and contextual features of stroke lesions. A weighted ensemble fusion strategy is used to improve robustness while maintaining efficiency. Furthermore, optimized preprocessing techniques such as intensity normalization, resampling, and spatial padding ensure consistency across scans [1]. Additionally, test-time augmentation (TTA) [6] is incorporated to enhance generalization without significantly increasing computational cost. The performance of the framework is evaluated using standard metrics such as Dice [6] and F1-scores [2] to ensure reliable performance under constrained resource settings. The contributions of this study are summarized as follows:

- This research proposed DSANet, a heterogeneous 3D ensemble that integrated a densely connected network, a residual segmentation network, and an attention-gated U-Net to capture complementary spatial and contextual features from FLAIR, DWI, and ADC sequences.
- We formulated an ensemble construction as a constrained grid-search problem that jointly optimizes fusion weights and the binarization threshold on

the validation set using a Dice-driven objective, eliminating the need for hand-tuned or uniform weighting schemes.
- We designed a resource-efficient training and inference pipeline that combines automatic mixed precision, gradient accumulation, test-time augmentation, and morphological post-processing, enabling accurate segmentation on a single GPU without high-end hardware.
- Lastly, we conducted extensive experiments on the ISLES 2022 dataset, achieving a macro Dice of 0.8780 and recall of 0.9279, and provide a subgroup analysis across lesion sizes and anatomical regions that demonstrates stable performance under diverse clinical conditions.

2 Related Work

De la Rosa et al. [5] established DeepISLES, an ensemble deep learning model that showed high accuracy in ischemic stroke lesion segmentation and a strong correlation with clinical assessment scores across a variety of real-world datasets. However, the model still faces challenges in accurately detecting small infarcts in regions prone to imaging artifacts or in cases involving complex imaging patterns from chronic lesions. Wu et al. [7] proposes a W-Net model for stroke lesion segmentation and achieves about 61.76% Dice score with improved boundary accuracy compared to existing methods. However, their model depends on complex design and does not focus on a simple and computationally efficient approach for practical use.

Zhang et al. [8] developed a universal foundation model called Mixture of Modality Experts (MoME) that utilizes specialized expert networks and a hierarchical gating system to automatically segment various brain lesion types across diverse MRI modalities. While this approach demonstrates high accuracy and strong generalization, it primarily relies on 3D volumetric data and may encounter computational constraints when attempting to integrate larger numbers of modality-specific experts or learning from extremely high-resolution context sets. Jazzar and Douik [9] developed the Res2U++ model using a multi-scale encoder and an ASPP block to achieve a training Dice score of 0.85 on the ISLES 2022 dataset. However, their architecture relies on complex feature extraction that requires high-end computing power, making it difficult to implement in low-resource countries where the necessary hardware is often unavailable.

Pacal et al. [10] achieved a high Dice score of 0.89 on the ISLES 2022 dataset by using gated convolutions and multi-scale attention to effectively capture both small and large stroke lesions. However, their model's reliance on 65.12 million parameters and specialized hardware for fast inference makes it difficult to deploy in low-resource countries where advanced computing power and high-end GPUs are often unavailable. Rahman et al. [11] introduced a multi-channel deep learning model using DenseNet121 and SelfONN to achieve a high Dice score of 87.49% on the ISLES 2022 dataset. However, their approach relies on expensive, high-end GPUs and the fusion of multiple MRI sequences, which are often unavailable in low-resource clinical settings. Relying only on single-modality data in these regions would significantly drop the model's performance, as it lacks the multi-spectral information needed to accurately distinguish stroke lesions from surrounding brain tissue.

Chu et al. [12] introduced StrokeFlow to improve boundary accuracy and the detection of small lesions on the ISLES 2022 dataset. However, the model requires specialized supervision and alignment with ADC maps, which can be difficult to implement in facilities where multi-modal data is inconsistent. Furthermore, the paradigm shift toward continuous field modeling involves complex mathematical computations that may not be sustainable in low-resource settings lacking advanced processing hardware. Existing models frequently achieve high accuracy by employing complex architectures that need large processing resources and costly GPUs. However, these models are challenging to implement in countries with limited resources where advanced technology and comprehensive multi-modal MRI sets are often unavailable. Moreover, these models operate as black-box systems, making their decision-making process difficult to interpret. As a result, there is a growing demand for efficient, single-network solutions that produce consistent results on conventional clinical equipment without the need for expensive infrastructure.

3 Methodology

The proposed method starts from multi-modal data acquisition and preprocessing to ensemble-based segmentation and evaluation. The overall working procedure consists of modality pairing, preprocessing, independent model training, ensemble fusion, post-processing, and quantitative assessment. Figure 1 presents the end-to-end pipeline for stroke lesion segmentation, covering preprocessing, multi-modal 3D input construction, multi-model training, and ensemble-based prediction and evaluation.

3.1 Data Preprocessing

This study uses the training set of the ISLES-2022[1] dataset [1] for experiments. Then we normalize the intensity values of all images and apply Z-score normalization [13], as shown in Eq. (1).

$$X_{\text{norm}} = \frac{X - \mu}{\sigma} \tag{1}$$

Here, μ is the mean intensity and σ is the standard deviation. To remove excess background noise and fit the GPU memory limits, this study applies spatial padding and random cropping [13]. The study resizes all volumes to a fixed dimension of $64 \times 64 \times 64$ and this uniform size ensures consistent model input. Furthermore, since medical datasets typically contain a limited amount of data, spatial augmentations are applied to prevent overfitting. Specifically, flips along the X, Y, and Z axes are performed using `RandFlipd` [14] with a 50% probability, and random 90-degree rotations are executed using `RandRotate90d` [14] with the same probability. Finally, to enhance memory efficiency and ensure rapid

[1] https://zenodo.org/records/7153326

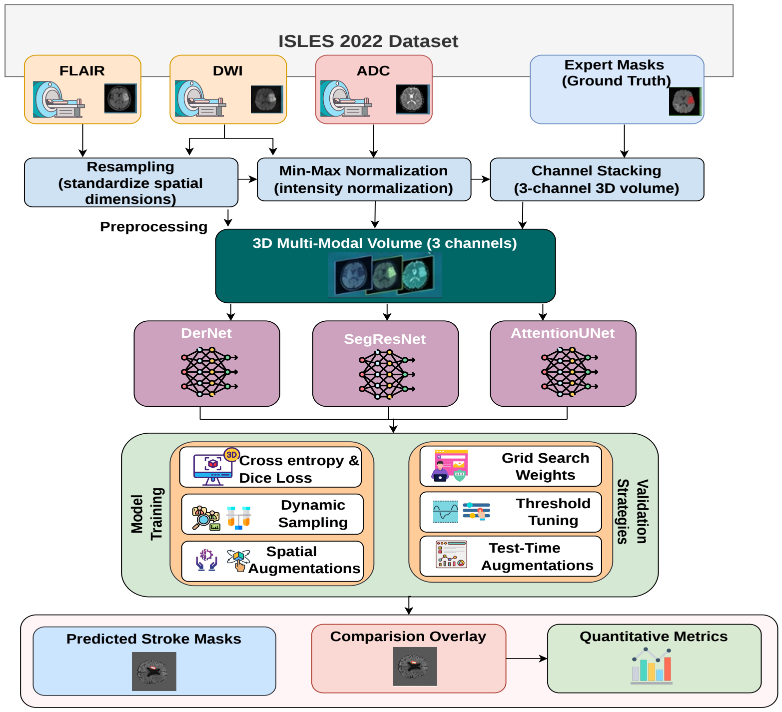

Fig. 1. Overview of the proposed DSANet framework. Preprocessed multi-modal MRI volumes (FLAIR, DWI, ADC) are processed in parallel by three complementary 3D networks-DerNet, SegResNet, and Attention U-Net. Their soft predictions are fused through weighted averaging, where the weights and binarization threshold are jointly selected by the constrained grid search in Algorithm 1, followed by morphological post-processing.

computations on the GPU, all data is converted to the `torch.float32` format using the `CastToTyped` function.

3.2 Problem Formulation

This study formulates stroke lesion segmentation as a supervised learning problem under resource-constrained settings. Let an input image be $X \in \mathbb{R}^{H\times W\times D\times C}$. Here, H, W, D represent the spatial dimensions and C represents the number of input modalities. The true label mask is $Y \in \{0,1\}^{H\times W\times D}$. The goal is to learn a mapping function f_θ that predicts a binary mask $\hat{Y}$, as shown in Eq. (2).

$$\hat{Y} = f_\theta(X) \tag{2}$$

θ represents the trainable model parameters. To ensure applicability in low-resource environments, this study emphasizes efficient learning with reduced computational complexity while maintaining segmentation accuracy. We optimize θ to minimize the difference between $\hat{Y}$ and Y.

3.3 Proposed DSANet Framework

We propose the DSANet framework for stroke lesion segmentation. The name stands for DerNet [15], SegResNet [16], and Attention U-Net [17] Network. DSANet uses a multi-branch architecture and extracts diverse features from the input images. The framework consists of three parallel branches. Each branch generates an independent prediction mask. An ensemble module combines these predictions. This combination improves the final segmentation accuracy. The DSANet consists of three distinct deep learning architectures. To ensure reproducibility and consistency across all experiments, all models were trained using a fixed random seed of 42. The input data was standardized into $64 \times 64 \times 64$ voxel crops. Data augmentation included random horizontal/vertical flips and rotations applied with a 50% probability to enhance model robustness. Optimization was performed using the AdamW optimizer [18] with a base learning rate of 1×10^{-4}. Furthermore, Automatic Mixed Precision (AMP) [19] was utilized to optimize memory efficiency and accelerate training speeds without compromising numerical stability. Finally, to maximize the evaluation accuracy on the test set, Test-Time Augmentation (TTA) [19] is applied.

DerNet. This study uses a DerNet model which connects each layer to every other layer. This structure ensures maximum information flow. The feature map of the l-th layer is x_l as shown in Eq. (3).

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}]) \tag{3}$$

H_l denotes a composite non-linear function. The brackets denote feature concatenation. DenseNet effectively extracts high-level semantic features. The study initializes the network using pre-trained weights and fine-tunes the model for 80 epochs.

SegResNet. This study uses the SegResNet model as the second model of the segmentation framework. SegResNet utilizes residual connections and a residual block computes as shown in Eq. (4):

$$y = F(x, W_i) + x \tag{4}$$

Here, x is the input and y is the output. $F(x, W_i)$ represents the residual mapping. These connections solve the vanishing gradient problem. SegResNet captures multi-scale contextual information. The network is initialized with specific architectural parameters, including an initial filter size of 32 and a dropout probability of 0.2. The study trains the SegResNet model for 100 epochs. To optimize memory usage while maintaining training stability, a gradient accumulation strategy is utilized.

Attention U-Net. The study uses the Attention U-Net model as the third model of the segmentation framework. The model incorporates attention gates to highlight salient features. These gates effectively suppress irrelevant background regions. The attention coefficient is α_i, as shown in Eq. (5).

$$\alpha_i = \sigma\left(W^T \,\mathrm{ReLU}(W_x x_i + W_g g_i + b_1) + b_2\right) \tag{5}$$

Here, x_i is the input feature, g_i is the gating signal, and σ is the sigmoid function. The Attention U-Net is trained for exactly 100 epochs. The specific hyperparameter configurations used for training the SegResNet, Attention U-Net and DerNet models are summarized in Table 1.

Table 1. Hyperparameter values for the segmentation models.

Parameter	**SegResNet**	**AttentionUNet**	**DerNet**
Input size	$64 \times 64 \times 64$	$64 \times 64 \times 64$	$64 \times 64 \times 64$
Batch size	2	2	2
Patch size	$64 \times 64 \times 64$	$64 \times 64 \times 64$	$64 \times 64 \times 64$
Epochs	100	100	80
Optimizer	AdamW	AdamW	AdamW
Initial Learning Rate	1e−4	1e−4	1e−4
Weight Decay	1e−5	1e−5	1e−5
Loss Function	DiceFocal Loss	DiceFocal Loss	DiceFocal Loss
Activation Function	ReLU	ReLU	LeakyReLU
Normalization	Instance Norm	Batch Norm	Instance Norm
Dropout Probability	0.2	–	–
Seed	42	42	42
AMP	True	True	True

3.4 Ensemble Strategy

This research proposes an advanced ensemble pipeline to maximize final prediction accuracy. First, TTA is applied to the test set. Each model processes the original image, an X-axis flipped version, and a Y-axis flipped version. The pipeline averages these three predictions. Next, an automated grid-search runs on the validation set to find the best combination weights for the three models. This search gives the highest weight priority to the DerNet model. Another grid-search finds the optimal threshold value. This step converts the weighted probability map into a final binary mask. Finally, a morphological post-processing step removes small, disconnected noise blobs under 30 pixels. This clean-up eliminates false positives and heavily refines the prediction. Here, this study combines the outputs of the three branches. Let the three individual predictions be P_1, P_2, P_3. We use a weighted averaging strategy. The final prediction is P_{final}, as shown in Eq. (6).

$$P_{\text{final}} = w_1 P_1 + w_2 P_2 + w_3 P_3 \tag{6}$$

Here, w_1, w_2, w_3 are learned weights and $w_1 + w_2 + w_3 = 1$. We apply a threshold T to P_{final}. This step creates the final binary mask. Algorithm 1 shows how the full procedure works.

Several deep learning models were tested to find the most effective way to segment stroke lesions. These models included SegResNet, Attention U-Net, DerNet, and DynUNet [20]. After the initial runs, DerNet, SegResNet, and Attention U-Net were selected due to their high accuracy and ability to capture lesion edges. These three models were combined into an ensemble to correct individual errors and reduce false results. Other models, such as DynUNet, were excluded because of high complexity and lower performance on the dataset presented in Table 2. The architectural diagram of DSANet is presented in Figure 2. The ensemble model is used to combine the unique architectural strengths of multiple models to achieve a more stable and balanced prediction. By merging the outputs of different networks, individual errors and biases are reduced, leading to more reliable segmentation across various lesion shapes and sizes.

Algorithm 1 Constrained Grid Search for Heterogeneous 3D Ensemble

Require: Validation dataset $D_{\text{val}} = \{(X_i, Y_i)\}_{i=1}^{N}$, Model pool $M = \{M_{\text{der}}, M_{\text{seg}}, M_{\text{att}}\}$, Candidate thresholds T, Lower-bound $\mu = 0.1$, Step size $\delta = 0.05$
Ensure: Optimal ensemble weights $\mathbf{w}^*$ and threshold t^*

Phase 1: Soft TTA Probability Extraction
for all $(X_i, Y_i) \in D_{\text{val}}$ **do**
  for all model $M_k \in M$ **do**
    $P_{k,i} \leftarrow \text{ExtractSoftTTA}(X_i, M_k)$ ▷ Before thresholding
  end for
end for
Phase 2: Constrained Grid Search
$S_{\text{best}} \leftarrow 0$
for $w_1 \leftarrow \mu$ **to** 1.0 **step** δ **do**
  for $w_2 \leftarrow \mu$ **to** $(1.0 - w_1)$ **step** δ **do**
    $w_3 \leftarrow 1.0 - (w_1 + w_2)$
    if $w_3 < \mu$ **then**
      continue
    end if
    for all $t \in T$ **do**
      $S_{\text{current}} \leftarrow 0$
      for all $(X_i, Y_i) \in D_{\text{val}}$ **do**
        $P_{\text{ens}} \leftarrow \sum_{k=1}^{3} w_k \cdot P_{k,i}$
        $\hat{Y}_i \leftarrow \mathbb{I}(P_{\text{ens}} > t)$
        $S_{\text{current}} \leftarrow S_{\text{current}} + \text{Dice}(\hat{Y}_i, Y_i)$
      end for
      if $(S_{\text{current}}/N) > S_{\text{best}}$ **then**
        $S_{\text{best}} \leftarrow (S_{\text{current}}/N)$
        $\mathbf{w}^* \leftarrow [w_1, w_2, w_3]$
        $t^* \leftarrow t$
      end if
    end for
  end for
end for
return $\mathbf{w}^*, t^*$

Table 2. Performance comparison of different deep learning models.

Model Name	Best Validation Dice	Final Test Dice
DerNet	0.8215	0.8136
SegResNet	0.7801	0.7819
Attention U-Net	0.7789	0.7274
DynUNet	0.6819	0.5796

3.5 Training Strategy and Low-Resource Learning Strategy

This study employs a robust training strategy. The loss function combines Dice Loss [21] and Focal Loss [22]. The total loss is L_{total} as shown in Eq. (7).

$$L_{\text{total}} = \lambda_1 L_{\text{Dice}} + \lambda_2 L_{\text{Focal}} \tag{7}$$

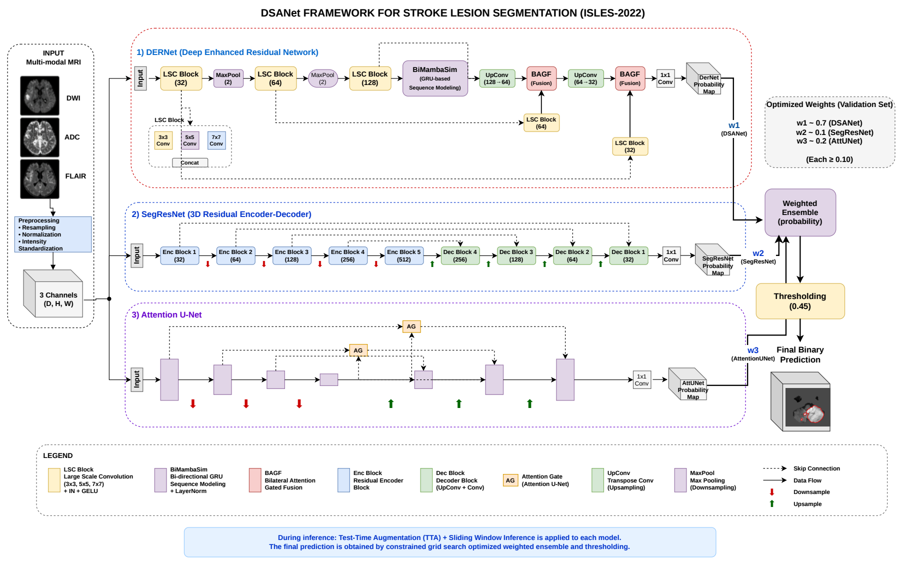

Fig. 2. The network follows an encoder and decoder structure where encoder features are selectively integrated into the decoder via BAGF gates, which control both spatial and channel-wise information flow instead of direct skip connections. A BiMamba-based context module captures global dependencies, while the gated fusion ensures that only lesion-relevant features are propagated, improving segmentation precision.

Here, λ_1 and λ_2 are balancing weights. Thus, this study reports the balancing weights of the hybrid loss function as $\lambda_1 = 1.0$ and $\lambda_2 = 1.0$, with a focusing parameter of $\gamma = 2.0$ for the Focal Loss component. The DiceFocalLoss effectively handles this imbalance problem. This study also uses gradient accumulation to simulate a larger batch size. A Cosine Annealing scheduler controls the learning rate. It smoothly decreases the learning rate during training. Medical image segmentation often has a huge class imbalance between the lesion and the background. The study uses AMP and a gradient scaler to speed up the training process and reduce GPU memory consumption. Hardware limitations prevent the use of large batch sizes. Therefore, the study applies a gradient accumulation technique. This technique ensures stable training by simulating a larger batch size. A sliding window inference is run on the validation set at the end of each epoch. The model weights with the highest F1-score are saved for final testing.

Medical image datasets generally contain a small amount of data. This study adopts a low-resource learning strategy to overcome this limitation. Several data augmentation techniques are applied during training. These techniques prevent the models from memorizing the training data. Next, random spatial crops are used during training. The study also applies random flips and 90-degree spatial rotations with a 50% probability. These augmentations help the models recognize new patterns. They significantly increase the generalization power on unseen data. For example, it generates an original version, an X-axis flipped version, and a Y-axis flipped version. The network processes all three versions independently. The pipeline then averages these three predictions. This specific method drastically boosts the final segmentation accuracy without requiring any extra training data.

3.6 Evaluation Protocol and Implementation Details

This study utilizes the 250 publicly available cases from the ISLES 2022 dataset for all experiments. Due to the unavailability of the hidden challenge test set, the data is partitioned into a fixed 70–15–15 split for training, validation, and testing, respectively. This study acknowledges that this custom split makes the results non-comparable to the official challenge leaderboard; however, it serves as a thorough internal evaluation framework. This study measures performance using three primary metrics. This study uses the Dice Similarity Coefficient (DSC). DSC measures spatial overlap, as shown in Eq. (8).

$$\mathrm{DSC} = \frac{2 \times |Y \cap \hat{Y}|}{|Y| + |\hat{Y}|} \tag{8}$$

Here, Y is the ground truth and $\hat{Y}$ is the prediction. This study also calculates Precision [23], Recall [23], and F1-score [23]. This study implements the framework using PyTorch [24] and the MONAI library [24] for medical image processing. This study runs the experiments on a single NVIDIA GPU [24] and uses mixed-precision training. This technique speeds up the training process. It reduces memory consumption.

4 Results

This section presents a comprehensive evaluation of the developed ensemble model on the ISLES-2022 dataset. First, the quantitative performance is assessed using standard medical image segmentation metrics to demonstrate the model's spatial accuracy and reliability. Following this, a qualitative analysis is provided to visually inspect the boundary delineation capabilities and network interpretability. Finally, the clinical relevance, architectural contributions, and the broader implications of the findings are discussed.

4.1 Quantitative Performance Evaluation

The performance of the DSANet-ISLES was evaluated on the test set. The final model utilized an optimized weighted ensemble strategy and an optimal cutoff threshold of 0.45, determined through an automated grid search on the validation set. The quantitative findings are presented using two complementing averaging methods: macro average (subject-wise mean) and micro average (global voxel-wise aggregate). The Mean Subject-wise Dice Score evaluates reliability across patients, whereas the global voxel-wise metrics give insights into overall pixel-level accuracy. The detailed performance metrics are presented in Table 3.

Table 3. Quantitative Segmentation Performance on the Test Set.

Metric (Averaging Method)	Mean Score (0 to 1)
Dice Similarity Coefficient (Macro)	0.8780
Intersection over Union (IoU – Macro)	0.7826
Precision (PPV – Macro)	0.8333
Recall (Sensitivity – Macro)	0.9279

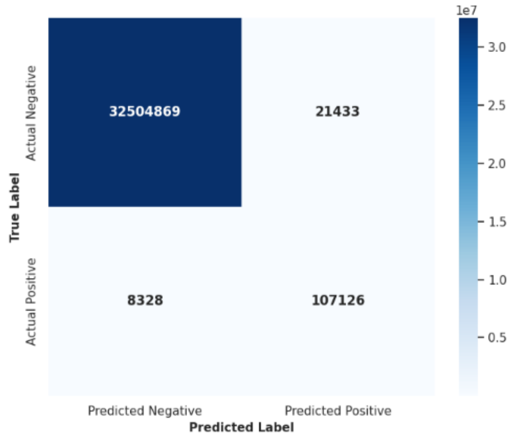

Fig. 3. Pixel-wise Confusion Matrix shows the model's performance in segmenting brain tumor regions compared to the ground truth.

To further analyze the pixel-level performance, a global confusion matrix was calculated. This matrix covers all test subjects and includes over 32.6 million voxels. The model achieved 107,126 true positives, 21,433 false positives, 8,328 false negatives, and 32,504,869 true negatives as shown in Figure 3. The micro F1-Score is 0.8780, which indicates a high overlap between the predicted stroke areas and the actual ground truth. Furthermore, the model achieved a micro Recall of 0.9279 which significantly reduces the risk of missing a critical lesion. Additionally, the micro Precision score of 0.8333 shows that when the model predicts a stroke, the prediction is correct 83.33% of the time. This study observes that the distribution of True Negatives relative to False Positives reflects the severe class imbalance, resulting in a remarkably low False Positive Rate of 6.6×10^{-4} and confirming the model's high specificity.

Table 4 shows that the ensemble achieves a Mean Dice of 0.9541 in deep brain structures and a near-perfect 0.9909 in the lesion core. While large lesions ($> 10\,\mathrm{ml}$) are segmented with high proficiency (0.8860), the model remains reliable for small, challenging infarcts with a score of 0.7306. These results confirm the framework's stability across diverse stroke sizes and anatomical locations, effectively outlining complex boundaries with a 0.8350 Dice score. These results demonstrate the framework's superior clinical reliability by maintaining high precision across diverse stroke sizes, anatomical locations, and complex lesion boundaries.

4.2 Qualitative Analysis and Boundary Analysis

A detailed qualitative analysis was also performed. This visual check evaluated the model's ability to outline lesion boundaries accurately. The data preprocessing pipeline was tested. The effect of intensity normalization on a raw DWI

Table 4. Performance Analysis across Lesion Sub-groups and Regions.

Category	Sub-group / Region	Mean Dice Score
Lesion Size	Small Lesions ($< 10\,\mathrm{ml}$)	0.7306
	Large Lesions ($> 10\,\mathrm{ml}$)	0.8860
Anatomical Location	Deep Brain Structures	0.9541
	Superficial Brain Regions	0.8037
Spatial Consistency	Lesion Core	0.9909
	Lesion Boundary	0.8350

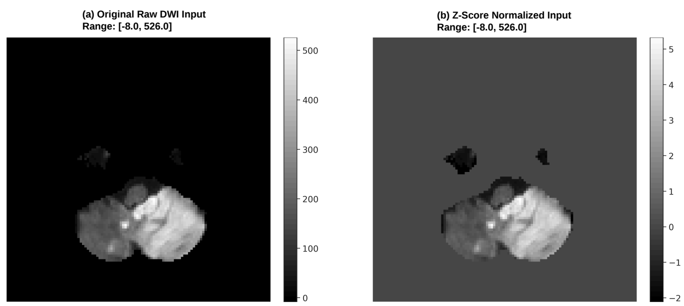

Fig. 4. Verification of the data pre-processing pipeline. Panel (a) shows the original raw DWI input with a high intensity range. Panel (b) displays the corresponding Z-score normalized input with a standardized intensity distribution.

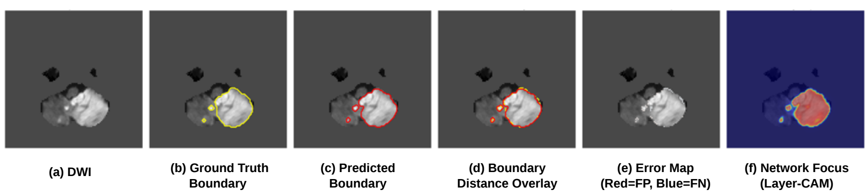

Fig. 5. Advanced Boundary and Focus Analysis on an Axial DWI slice. (a) Input DWI; (b) Ground truth lesion boundary (yellow contour); (c) Predicted lesion boundary (red contour); (d) Boundary Distance Overlay comparing overlap; (e) Error Map: offers a detailed perspective of the segmentation quality, the minimal presence of red and blue pixels confirms the model's ability to minimize both over-segmentation and missed detections; (f) The Layer-CAM visualization serves as explainable AI (XAI) evidence, proving that the network focuses its attention specifically on the highly intense ischemic regions rather than being distracted by noise or healthy anatomical structures.

input was visualized. Figure 4 compares the raw DWI image with its Z-score [13] normalized version. The normalized image shows better contrast. This ensures that intensity changes do not reduce the segmentation accuracy. An advanced 6-panel boundary analysis grid was created. This grid compares the model's predictions with the expert ground truth. An example of an axial slice with stroke lesions is shown in Figure 5. Figure 6 shows the qualitative segmentation results, where the red contours represent the predicted stroke lesions. It demonstrates the model's ability to accurately identify multifocal and irregular ischemic regions across different brain slices.

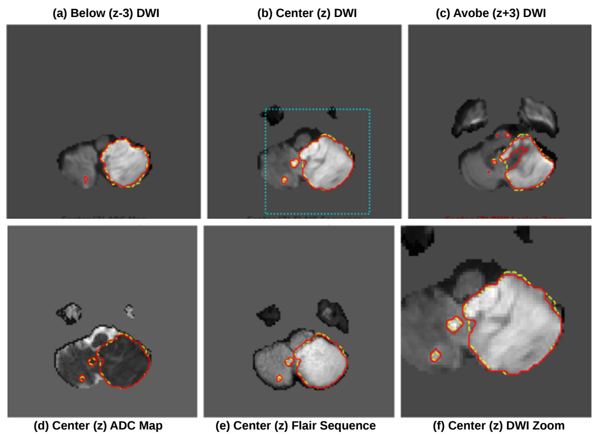

Fig. 6. Qualitative segmentation results show the model's robustness across different anatomical planes: (a) Below (Z-3), (b) Center (Z), and (c) Above (Z+3), capturing the full 3D extent of the lesion. The consistent red contouring across the (d) ADC maps (indicating restricted diffusion) and (e) FLAIR images shows the model's predictions are comparable to ground truth. Furthermore, the (f) The DWI lesion zoom highlights the model's ability to preserve detailed lesion boundaries.

Figure 5(e) shows the spatial error distribution, reflecting 21,433 false positive and 8,328 false negative voxels across the test set. The higher count of false positives indicates a slight tendency of the model to over-segment healthy areas around the lesion boundaries. However, the visually minimal false negatives align with the high Recall (107,126 true positives), showing that the model rarely misses critical stroke areas, thereby ensuring high clinical safety.

4.3 Discussion

The results show that the proposed DSANet-ISLES model performs very well on the ISLES-2022 dataset. The macro-average Dice score and micro-average F1 score are closely aligned, indicating consistent and reliable segmentation performance across the dataset. It also confirms the framework's effectiveness for both large and small lesions. This balance ensures clinical robustness across varying stroke severities. In clinical settings, maintaining high precision is essential to avoid unnecessary incorrect predictions. The model shows a very low number of false positives relative to true negatives, demonstrating strong reliability in distinguishing healthy brain tissue from stroke regions. The automated weight optimization also gave interesting results. It assigned a high weight (0.70) to DerNet. Attention U-Net received a smaller weight (0.20), and SegResNet received the lowest (0.10). While DerNet provides the core segmentation performance, the inclusion of Attention U-Net and SegResNet is vital for ensuring the framework's robustness and generalizability. Specifically, incorporating these models provides architectural diversity, allowing the ensemble to leverage multi-scale features and attention mechanisms that a single model might overlook. Furthermore, this multi-model integration helps in refining complex lesion boundaries and reducing false negatives, ensuring more stable performance across the high morphological variability of stroke lesions in clinical settings. In the future, the model can be improved further. Small lesions with blurry boundaries are still hard to detect accurately. Future work will add edge-aware loss functions to solve this problem.

5 Conclusion

This study addressed the challenge of deploying accurate ischemic stroke lesion segmentation in computationally constrained environments, where heavy architectures and specialized hardware are often unavailable. We proposed DSANet, a heterogeneous ensemble of three complementary 3D architectures whose predictions are fused through a constrained grid-search that jointly optimizes fusion weights and the binarization threshold on the validation set. The framework demonstrates that architectural diversity, when coupled with a principled optimization of the fusion stage, can outperform individual high-capacity models without incurring their computational cost, and subgroup analysis further indicates stable behavior across lesion sizes and anatomical regions. Nevertheless, certain limitations remain: small infarcts and blurry lesion boundaries continue to present the greatest difficulty, and the reliance on the full multi-modal input may limit applicability in settings where not all MRI sequences are available. Future work will incorporate boundary-aware loss formulations to improve delineation of small and irregular lesions, explore modality-dropout training strategies to handle missing sequences gracefully, and validate the framework on external multi-center cohorts to assess robustness under diverse acquisition protocols.

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